10 extraction and 2 wash stages, what percentage of (b) uranium and (c) caesium is extracted? In an extraction equipment D_{Cs} 0.003 cannot be maintained, because droplets are carried over between stages; the practical value will be D'_{Cs} 0.02. (d) How much caesium is extracted in this latter case with the counter current equipment? Assume equal phase volumes.

A.2. After scrubbing the solvent (1M TBP in kerosene) with a carbonate solution it should be reacidified to equilibrium with 6 M nitric acid before reuse in reprocessing operations. Assume that the partial molar volume of nitric acid can be neglected for both the organic phase and the aqueous phase. Thus, in this example extraction of nitric acid will not increase the volume of the organic phase nor reduce the volume of the aqueous phase. Extraction of nitric acid is assumed to only occur as the complex $TBP \cdot HNO_3$. For simplicity we will assume that the Eqn $D_{HNO_3} = k_{ex} \times [HNO_3]_{aq} \times [TBP]_{free,org}$, with $k_{ex} = 0.1$ can be used to calculate the D-value for nitric acid. The reacidification will be performed in a simple counter current mixer-settler battery. (a) How many stages and (b) what flow rate ratio (org/aq) would be needed in order to produce the required nitric acid concentration in the organic phase using the smallest possible feed of 6.4 M nitric acid? (c) What would be the concentration of the spent nitric acid?

A.6. LITERATURE

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APPENDIX B

Answers to Exercises

Chapter 2: (1) π-mesons, conversion of excess energy into mass. **(2a)** gravitation, the electromagnetic force, the weak interaction force, and the strong interaction force. (**2b**) See e.g. Tables 2.2 and 2.3. **(3a)** Bosons are a group of 'particles' to which the photon and mesons belong. The bosons are the carriers of forces. When two fermions interact they continually emit and absorb bosons. The bosons have an even spin (0, 1, etc.), they do not obey the Pauli principle, and they do not require the formation of antiparticles in their reactions. Particles of nonintegral spin are called fermions because they obey the statistical rules devised by Fermi and Dirac, which state that two such particles cannot exist in the same closed system (nucleus or electron shell) having all quantum numbers the same (referred to as the Pauli principle). Fermions are the building blocks of nature. Fermions can be created and destroyed only in conjunction with an antiparticle of the same class. For example if an electron is emitted in β-decay it must be accompanied by the creation of an antineutrino. Conversely, if a positron − which is an antielectron − is emitted in the β-decay, it is accompanied by the creation of a neutrino. (**3b**) Answer is included in, and follows from, answer to (a). **(4)** Radiation pressure, recoil in photon and γ-emission, energy increase and decrease when falling into or moving out against a gravity field. **(5)** Several possibilities, e.g.: Reaction with ^{1}H to produce a fast neutron and a positron. Then detection of annihilation γ and neutron, $^{37}Cl + n = {}^{37}Ar + e^{-}$ followed by collection of Ar and detection of its decay, $^{71}Ga + n = {}^{71}Ge + e^{-}$, conversion to GeH_4, collection and detection of its decay.

Chapter 3: (1) 1.553×10^{22}. **(2)** 0.100. **(3a)** 2111 m s^{-1}. **(3b)** 0.0462 V **(4)** 472 V **(5)** 11.00934 u. **(6)** 0.129 nm **(7a)** 3.032×10^{17}. (**7b**) 3322 K **(8)** 2.080. **(9)** 4. (**10**) 0.030 m (**11a)** By series expansion $e^{\delta} = 1 + \delta + \delta^2/2 + \ldots$but $\alpha = 1 + \delta$ hence $e^{\delta} \simeq \alpha$. (**11b)** 1.097. (**11c)** 22. (**12)** 5.742 SWU.

Chapter 4: (1) 8.261 MeV **(2)** 1.581×10^{6}. (**3**) 47371 km (**4**) Fusion gives 7 times more per gram. (**5**) −1.02 MeV, −2.76 MeV, and −1.04 MeV (**6**) 6.262 MeV u^{-1}, 8.366 MeV u^{-1}, 8.734 MeV u^{-1}, and 7.489 MeV u^{-1}. (**7**) 4.8, 12.6, 26.4, and 91.8.

Chapter 5: (1) 0.088 MeV, ^{235}U. **(2a)** 0.87 MeV **(2b)** 0.46 MeV (measured 0.51 MeV). **(3)** 99 eV **(4)** 1.26 keV **(5)** 547 keV (measured 662 keV). **(6)**. 1.28×10^{9} years **(7)** 5.9×10^{9} years **(8)** 2.21×10^{22} J $year^{-1}$. **(9a)** 16.9 GBq. **(9b)** 3.0×10^{-10} m^3 **(10a)** Multiply (5.25) by $\mathbf{c}^2$. **(10b)** $m_{\gamma}^{0} = 0 \rightarrow m_{\gamma} = E_{\gamma}/\mathbf{c}^2$ hence $E_{\gamma}(E_{\gamma}/\mathbf{c}^2) = E_{d}m_{d}$. **(11a)** 1.46×10^{4} Bq. **(11b)** 8.09×10^{8} atoms. **(12)** $t_{½} = 104$ h and 99 Bq, $t_{½}$ 4.4 h and 1030 Bq. **(13)** R_0 507 cpm for $t_{½}$ 107 min, R_0 440 cpm for $t_{½}$ 24 min (cf. ^{298}At $t_{½}$ 98 min, ^{206}At $t_{½}$ 31 min) **(14)** 2.3×10^{5} years.